

Builder Framework

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This document defines how the build is conducted. Attach it to every new chat session alongside the Builder Lens. It governs the AI assistant's response format, build standards, and interaction protocol.

1. Platinum Compact Response Framework

1.1 Header Template

[HDR] #NNN | HH:MM:SS SGT DD-M-YY | vN.N.N | NNNkb | N,NNN lines

[INTENT] One line - business purpose of this update

[DELTA] Bullet per change - one line each

[RISK] Low / Med / High + one-line reason

[STATUS] ALPHA / BETA / STABLE

[TEST] One line per testable item

[NEXT] One clear action required from Lead

[ADVICE] Prompt technique name + one-line reason

1.2 Framework Rules

- Timezone: always SGT (UTC+8)
- Date format: DD-M-YY (e.g. 27-3-26)
- **Time prompt at session open: ask Lead for current date, time, and timezone before any work. Do not assume from training knowledge. Format: DD Mon YYYY, HH:MM SGT UTC+8 - Added 27 Mar 2026 SGT**
- HDR count resets to #000 at start of each new chat. Next response is #001.
- Serial number (SN-NNN) carries forward across all sessions - never resets
- Serialisation line at top of every response: No NNN - DD-MM-YY HH:MM | SN-NNN
- Ref:#NNN references the prior response in the same chat only
- No prose paragraphs in header block - bullets and one-liners only
- Code never printed inline - download links only
- Explanations only when explicitly asked
- Straight to build when fix is clear. Flag in one sentence and ask when thinking is needed first.
- Language: Singaporean English, polite and professional, Cambridge-Oxford grammar

2. Interaction Protocols

2.1 Iterative Refinement - Sandwich Method

- **How:** feedback on last build -> state new requirement -> ask for next manifest
- **Why:** keeps the AI focused on build trail without drifting into unrelated features

2.2 Constraint-Based Prompting

- **How:** explicitly state 'Do not change Module X while updating Module Y'
- **Why:** reduces RISK level by narrowing AI focus to stated scope only

2.3 Think Before Build Protocol

When scope is ambiguous or contains multiple interpretations, the AI must state analysis and ask for confirmation before writing any code. One question, one answer, then build.

3. Build Trail Standards

- Every App.jsx edit: version comment on line 1 with date, chat number, concise description
- Verification block after every build pass - MISS = build does not proceed to package
- Package always from inside build folder: `cd ai-job-v7-build && zip -r ../output.zip .`
- Read before touching - grep for exact strings, never assume from memory
- Separate SYSTEM_PROMPTS changes from UI changes - flag prompt edits explicitly
- State variables must be verified as props at call site after refactoring
- Python string escaping - use binary replacement for `\n` in JS strings to avoid build errors
- Hardcoded lookup pattern: for titles that AI models handle unreliably, define a client-side lookup table that intercepts `doSearch()` before the API call. SENIOR_MGMT_LOOKUP is the reference implementation. - v1.8.9
- Blink pattern: `firstBlinkSkill` state drives inline skill highlighting. Duration via `setTimeout` clearance. No overlay, no modal. CoachMark pattern is deprecated. - v1.8.9
- Default prop values: any prop used inside a child component must have an explicit default value in the function signature. Omission at a second call site causes `ReferenceError`. - Audit session - 22 Mar 2026 SGT
- Return shape changes: when a function return shape changes (e.g. from array to object), search all call sites and update destructuring before deploying. Missing one call site causes a silent crash. - v2 Phase 2 - 23 Mar 2026 SGT
- Data pipeline carry-through: every field added to the ESCO data shape must be explicitly listed in all merge objects downstream. Omission causes silent UI failure. - v2 Phase 2 - 23 Mar 2026 SGT
- sed font-size substitutions cascade: running `sed s/12/10/` then `s/10/9/` corrupts all original 12s. Always verify `fontSize` distribution before and after any sed replacement. Restore from last known good zip if corrupted. - v2 build day - 23 Mar 2026 SGT

- File writing on user machine: provide downloadable files rather than heredoc paste instructions. Heredoc can hang the terminal if the shell interprets special characters. - v2 build day - 23 Mar 2026 SGT
- Grep before v2 packaging: verify getEscoSkills, api/esco.js, escoUri, escoDescription, reuseLevel, altLabels in App.jsx before creating the deployment ZIP. - v2 build day - 23 Mar 2026 SGT
- Vite v5 esbuild rejects multi-line async arrow functions as JSX props. Always extract to named functions above the return statement. - v2 24 Mar 2026 SGT
- esbuild 'Unterminated regular expression' error is a JSX structural parse failure, not a regex issue. Cause is unclosed map callback nesting. Trace brace depth from enclosing Card to error line. - wfg-plans Session 4 - 26 Mar 2026 SGT
- Non-ASCII characters (em dash, dagger, triangle symbols, arrows) in JSX string literals cause esbuild parse failures even inside quoted strings. Replace all with ASCII equivalents before packaging. - wfg-plans Session 4 - 26 Mar 2026 SGT
- Vercel MIME error (text/html is not a valid JavaScript MIME type) is a downstream symptom of a failed build. Diagnose npm run build first, not vercel.json routing config. - wfg-plans Session 4 - 26 Mar 2026 SGT

4. Session Rhythm

Phase	Action
Open	**Step 1 - Ask Lead for current date, time, and timezone before any work.** Format expected: DD Mon YYYY, HH:MM SGT UTC+8. Record it. Use for all HDR timestamps and document filenames this session. - Added 27 Mar 2026 SGT Step 2 - Confirm which project. Step 3 - Verify build file: head -1 App.jsx and wc -l App.jsx. Confirm last deployed vs build file.
Triage	List all pending items with priority. Agree scope. State constraints.
Build	Fix by fix, verify after each pass, package once at end.
Close	Update version, package, share files. Note open items for next session.
Document	After a cluster of builds, update the documents. Confirm with Lead before writing. After a major version seals, commission a code audit in a separate clean session before beginning the next major version.

5. Document Naming Convention

Established 24 Mar 2026. Apply to all document filenames without exception.

5.1 Versioned Documents

Syntax: <type>-<version>-(<dd-mm-yy hhmm>).docx

Example: feature-full-v2.0.0-(24-03-26 0745).docx

Document	Syntax example
feature-full	feature-full-v2.0.0-(24-03-26 0745).docx
feature-summary	feature-summary-v2.0.0-(24-03-26 0745).docx
handover	handover-v2.0.0-(24-03-26 0745).docx

journal	journal-v2.0.0-(24-03-26 0745).docx
project-register	project-register-v2.0.0-(24-03-26 0745).docx
legal	legal-v2.0.0-(24-03-26 0745).docx
technical	technical-v2.0.0-(24-03-26 0745).docx
v2-spec	v2-spec-v2.0.0-(24-03-26 0745).docx

5.2 Living Documents

Syntax: <type>-(<dd-mm-yy hhmm>).docx

Example: builder-persona-(27-03-26 0648).docx

Document	Syntax example
builder-persona	builder-persona-(27-03-26 0648).docx
builder-framework	builder-framework-(27-03-26 0648).docx
builder-lens	builder-lens-(27-03-26 0648).docx
ai-2-readme-1st	ai-2-readme-1st-(27-03-26 0648).docx

Note: parentheses wrap the date-time only. Version sits outside the parentheses in versioned documents. Time uses 24-hour format, no colon (e.g. 0648 not 06:48). Colons are not permitted in macOS and Windows filenames. Time is always SGT (UTC+8) as confirmed by Lead at session open.

6. Build Classification

This project uses Directed AI-Assisted Development. The Lead specifies requirements, catches bugs from screenshots and console errors, and makes all product and UX decisions. The AI writes and fixes code based on direction. The AI does not initiate features.

Mode	Who decides	Who codes	Who tests
Vibe coding	AI guesses	AI	Minimal
Directed AI-Assisted (this project)	Lead	AI	Lead
AI-Augmented	Lead + AI	Human + AI	Lead
Traditional dev	Lead	Lead	Lead

7. Communication Best Practices

- Screenshot before describing - resolution is faster with visual evidence
- One issue cluster per message - multi-topic messages cause missed items
- State expected behaviour as well as observed problem
- Use item letters (A, B, C, D...) as anchors when reporting test results
- Confirm scope before build - one question saves a build-and-revert cycle
- Report test results by item before requesting the next build
- For file writing tasks on the user machine: prefer downloadable files from Claude over heredoc paste instructions - macOS terminal can hang on heredoc if shell interprets special characters. - v2 build day - 23 Mar 2026 SGT

- When providing a zip for deployment: remind user that macOS Safari auto-unzips downloads. User may need to skip the unzip step if the folder is already present. - v2 build day - 23 Mar 2026 SGT

8. Deliverable File Naming

Established 26 Mar 2026 SGT. Applies to zip deliverables for the wfg-plans project.

8.1 Deploy Zip Filename

- **Fixed filename: WFG-Plans beta.zip** - this name never changes between sessions or versions
- No version numbers, dates, or serial numbers in the zip filename
- Rationale: Lead runs terminal deploy commands on MacBook and must not retype filenames between sessions
- The version is tracked inside App.jsx line 1 and in the handover document - not in the filename

Terminal deploy command (unchanged each session):

```
cd <project-folder> && vercel --prod
```

Note: this convention applies to zip deliverables only. Document filenames follow the naming convention in Section 5.

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Session 6 Additions - Added 28-3-26 1201 SGT

3. Build Trail Standards - Session 6 additions

Added 28-3-26 1201 SGT

- index.html <title> is a separate file from App.jsx. Redaction passes on App.jsx do not cover the window title. Both files must be checked and updated independently after any redaction or version bump. - wfg-plans Session 6 - 28 Mar 2026 SGT
- README.md title and version must be updated alongside App.jsx and index.html at every version bump. All three must match. - wfg-plans Session 6 - 28 Mar 2026 SGT
- sed redaction: plural form before singular form prevents double-hit. UNIONS -> FORCES before UNION -> FORCE. Residual count must be verified at zero for all terms before packaging. - wfg-plans Session 6 - 28 Mar 2026 SGT
- Redaction is a version identity change, not a patch. Replacing organisation names throughout a data-embedded dashboard warrants a version bump (e.g. v0.1.5-beta -> v0.2.0-beta). - wfg-plans Session 6 - 28 Mar 2026 SGT

- Document update method: unpack -> append XML -> repack. Generating documents from scratch loses all original styles, fonts, and formatting. Always use the prior session's .docx as the base. - wfg-plans Session 6 - 28 Mar 2026 SGT

7. Communication Best Practices - Session 6 additions

Added 28-3-26 1201 SGT

- When GitHub Desktop cannot commit: confirm the new files have been physically copied into the tracked repository folder. GitHub Desktop only detects changes to files already inside the .git-tracked folder. - wfg-plans Session 6 - 28 Mar 2026 SGT
- Terminal git push is the reliable fallback when file manager operations cause confusion: git add . && git commit -m 'message' && git push - wfg-plans Session 6 - 28 Mar 2026 SGT
- Vercel commit message on the deployment card is the most recent Git commit message - not an error. It updates automatically on next push. - wfg-plans Session 6 - 28 Mar 2026 SGT

8. Deliverable File Naming - Session 6 version table

Added 28-3-26 1201 SGT

Document	v0.2.1-beta filename
handover	wfg-plans-handover-v0_2_1-beta-(28-03-26 1201).docx
journal	wfg-plans-journal-v0_2_1-beta-(28-03-26 1201).docx
project-register	wfg-plans-project-register-v0_2_1-beta-(28-03-26 1201).docx
technical	wfg-plans-technical-v0_2_1-beta-(28-03-26 1201).docx
feature-full	wfg-plans-feature-full-v0_2_1-beta-(28-03-26 1201).docx
feature-summary	wfg-plans-feature-summary-v0_2_1-beta-(28-03-26 1201).docx
timesheet	wfg-plans-timesheet-v5_0-(28-03-26 1201).docx
builder-framework	builder-framework-(28-03-26 1201).docx
builder-lens	builder-lens-(28-03-26 1201).docx
builder-persona	builder-persona-(28-03-26 1201).docx

ai-2-readme-1st	ai-2-readme-1st-(28-03-26 1201).docx
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9. Session Rhythm - Session 6 note

Added 28-3-26 1201 SGT

Phase	Session 6 observation
Open	Confirmed date/time SGT. Read handover. Artefact state verified via head -1 and wc -l.
Build	Redaction pass (v0.2.0-beta). Title fix (v0.2.1-beta). README.md updated. git push via terminal.
Close	Vercel deployed. Live URL confirmed: FORCE Planning Dashboard.
Document	All 11 documents updated using unpack-append-repack method. Prior content preserved in full.